A Point-Like Dark-Matter Target at 12.8 GHz from a QCD-Anchored Vacuum Condensate: Theory and Haloscope Test Protocol

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https://github.com/infosave2007/vmf

August 5, 2026

Abstract

The Null-Vector Gravity (NVG) framework, anchored to a single QCD input $M_{\Omega,0} = 859 \pm 8$ MeV fixed by lattice sigma terms, contains a pseudo-Goldstone mode θ of the vacuum condensate $\Phi = We^{i\theta}$. We examine the phenomenological consequences of the hypothesis that the same decay constant f_a that sets the θ-mode mass also generates Majorana neutrino masses through a Weinberg-like dimension-five operator suppressed by a two-loop QCD-anomaly coefficient. Introducing the unknown overall coefficient of this operator as an explicit parameter c_ν, the benchmark choice $c_\nu = 1$ fixes $f_a = 1.07 \times 10^{11}$ GeV from the neutrino sector and yields a *conditional point target* for dark matter: a coherent pseudo-scalar of mass $m_\theta = 53.2$ μeV, detectable as a microwave line at $f_\theta = 12.86$ GHz with photon coupling $g_{\theta\gamma\gamma} = 2.08 \times 10^{-14}$ GeV^{-1} (KSVZ benchmark). The product $m_\theta f_a$ agrees with the QCD axion band at 2.6% — a normalization cross-check, not an independent test — and the benchmark frequency lies in the currently unexplored gap between the published ORGAN mass windows (25–26 and 63–67 μeV), directly on the lower edge of the SMASH band (50–200 μeV) that ORGAN has declared as its roadmap target. We present the derivation, a calibration of the haloscope signal formula against the published ORGAN signal estimate, a Monte Carlo propagation of the statistical input uncertainties (m_3, $\chi_{\rm top}$), together with explicit conditional sensitivity scenarios for the unknown coefficient c_ν, and two concrete test protocols: (A) a resonant-cavity array and (B) a *fixed-position* dielectric booster stack, where the point target eliminates the scanning mechanism. For a coherent dark-matter fraction of 24% (misalignment only), protocol B reaches SNR = 5 in $\sim$18 h at power boost $\beta^2 = 10^4$ (MADMAX convention, $\beta^2 \equiv P_{\rm sig}/P_0$). Discovery and exclusion criteria are preregistered. This is a *phenomenological target paper*: the θ-seesaw operator is stated as a hypothesis, its coefficient is propagated as a parameter, and every derived quantity is conditional on it. All numerical values are reproducible from the accompanying scripts.

Keywords: axion dark matter, haloscope, dielectric haloscope, pseudo-Goldstone boson, vacuum condensate, neutrino mass, phenomenological target

Contents

1 Introduction

Direct searches for QCD axion dark matter have so far produced exclusion limits in disjoint mass windows: ADMX covers ~ 2.7–4.2 μeV [14], HAYSTAC covers ~ 17–20 μeV [15], ORGAN has scanned 25.45–26.27 μeV [18], 63.2–67.1 μeV [16] and 107.42–111.93 μeV [17], while MADMAX is under construction for the 40–400 μeV range [21]. The intermediate region around 50–60 μeV (12–15 GHz) remains unexplored.

This note reports that the NVG/VMF framework [2], anchored to a single QCD input [1], yields — conditional on one stated hypothesis (the θ-seesaw, Sec. 2.3) — a point target *exactly* in that gap: a dark-matter pseudo-scalar at 53.2 μeV. Because the target is a *point* in (m, g_γ) space rather than a band, the experimental problem simplifies drastically: no scanning, no tuning mechanism — a fixed-frequency integration.

Section 2 gives the derivation. Section 3 states the observable signature. Section 4 documents the gap in published searches. Section 5 defines the test protocol, including a calibration of the signal formula against published data and preregistered discovery/exclusion criteria. Section 6 lists the framework assumptions honestly.

2 Theoretical Framework

2.1 The anchor: $M_{\Omega,0}$ from lattice sigma terms

The framework is anchored to one dimensionful QCD input. Separating the nucleon mass into current-quark and nonperturbative components via the Feynman–Hellmann sigma terms ($\sigma_{\pi N} \approx 44$ MeV, $\sigma_{sN} \approx 30$ MeV), the vacuum value of the nonperturbative component is fixed at [1]

$$M_{\Omega,0} = 859 \pm 8 \text{ MeV}, \tag{1}$$

which sets the condensate energy density scale $\rho_c \approx 7.09 \times 10^4$ MeV/fm^3 used throughout the construction. No cosmological or astrophysical input is used at any stage.

2.2 The θ pseudo-Goldstone mode

The NVG vacuum condensate is parametrized as $\Phi = W\, e^{i\theta}$, where the phase θ is a pseudo-Goldstone mode with the standard anomaly-generated potential. Its mass obeys the same relation as the QCD axion,

$$m_\theta = \frac{\sqrt{\chi_{\rm top}}}{f_a}, \tag{2}$$

with the topological susceptibility $\chi_{\rm top}^{1/4} = 75.5$ MeV from lattice QCD [10]. In the normalization $m_a(f_a = 10^{12}\ \text{GeV}) = 5.691\ \mu\text{eV}$ [10], this becomes $m_\theta = 5.691\ \mu\text{eV} \times (10^{12}\ \text{GeV}/f_a)$.

2.3 The θ-seesaw: fixing f_a from the neutrino sector

The framework hypothesis behind the target is the *θ-seesaw*: the same mode θ generates Majorana neutrino masses without right-handed neutrinos. To be consistent with the electroweak gauge group above EWSB, the operator must have the unique gauge-invariant dimension-five (Weinberg) structure [7]; the NVG hypothesis specifies only its suppression scale and coefficient,

$$\mathcal{L}_{\rm eff} \;=\; 2\,c_\nu \left(\frac{\alpha_s}{4\pi}\right)^2 \frac{1}{f_a}\,(\tilde{H}^\dagger L)^T C\,(\tilde{H}^\dagger L) + \text{h.c.}, \qquad \tilde{H} \equiv i\sigma_2 H^*, \tag{3}$$

where c_ν is an unknown dimensionless coefficient. The normalization convention of this Weinberg-like operator, including symmetry factors in flavor space and the factors of $1/\sqrt{2}$ from $\langle H\rangle = v_{\rm EW}/\sqrt{2}$, is absorbed into c_ν; accordingly, $c_\nu = 1$ defines a benchmark normalization rather than a UV-derived coefficient. The operator is *defined* through the physical post-EWSB relation, for the heaviest neutrino,

$$m_3 = c_\nu \left(\frac{\alpha_s}{4\pi}\right)^2 \frac{v_{\rm EW}^2}{f_a}. \tag{4}$$

With $\alpha_s(M_Z) = 0.1184$, $v_{\rm EW} = 246.22$ GeV and the oscillation-and-cosmology anchor $m_3 = 50.3$ meV (normal ordering; the implied sum $\sum m_\nu = 59$ meV passes the DESI DR2 bound $\sum m_\nu < 64$ meV unconditionally), Eq. (4) fixes

$$\boxed{f_a = c_\nu \left(\frac{\alpha_s}{4\pi}\right)^2 \frac{v_{\rm EW}^2}{m_3} = 1.07 \times 10^{11}\, c_\nu\ \text{GeV}} \tag{5}$$

We stress the status of Eq. (3): it is a *framework hypothesis*, the single non-standard ingredient of this note. The nominal coefficient $(\alpha_s/4\pi)^2 \approx 8.9 \times 10^{-5}$ is motivated by the two-loop QCD-anomaly diagram, but possible flavor factors and RG running are not computed here, and the UV completion that generates the operator is not derived. The parameter c_ν absorbs all of this freedom: the benchmark $c_\nu = 1$ defines the headline target, and every result below is rescaled with c_ν (Section 5.4). The haloscope test discriminates the hypothesis directly (Section 6).

2.4 Point target for the dark-matter mass

Because every derived quantity depends on c_ν through f_a alone, the parametric structure is transparent (for fixed E/N):

$$f_a = f_{a,0}\,c_\nu, \qquad m_\theta = \frac{m_{\theta,0}}{c_\nu}, \qquad f_\theta = \frac{f_{\theta,0}}{c_\nu}, \qquad g_{\theta\gamma\gamma} = \frac{g_{\theta\gamma\gamma,0}}{c_\nu}, \tag{6}$$

where the subscript 0 denotes the benchmark values at $c_\nu = 1$. A larger c_ν means a larger f_a, hence a *smaller* mass and frequency and a weaker photon coupling. Inserting (5) into (2):

$$\boxed{m_\theta = 5.691\ \mu\text{eV} \times \frac{10^{12}}{1.07 \times 10^{11}\, c_\nu} = \frac{53.2}{c_\nu}\ \mu\text{eV}} \tag{7}$$

or, equivalently, a Compton frequency

$$f_\theta = \frac{m_\theta c^2}{h} = \frac{12.86}{c_\nu}\ \text{GHz}. \tag{8}$$

The benchmark $c_\nu = 1$ gives $m_\theta = 53.2\ \mu$eV, $f_\theta = 12.86$ GHz throughout the rest of this note.

Point versus band. If a UV completion fixes c_ν, the target is a genuine *conditional point* in (m, g_γ, f) space. If c_ν remains free, the point degrades to a *broad conditional search band* along the one-parameter curve (6) — 4.3–38.6 GHz for $c_\nu \in [1/3, 3]$ (Sec. 5.4) — and the fixed-frequency advantage is retained only conditionally, at a chosen benchmark value of c_ν. We use the terminology "conditional point target" and "broad conditional search band" accordingly.

Normalization cross-check (not an independent test). Once Eq. (2) is accepted, the product $m_\theta f_a = \sqrt{\chi_{\rm top}}$ is fixed by the same QCD input, so its landing on the QCD axion band $m_a f_a \simeq m_\pi f_\pi \sqrt{z}/(1+z) \approx 5.82 \times 10^{15}$ eV^2 [10] cannot count as an independent confirmation of NVG or of the seesaw hypothesis. The framework gives $m_\theta f_a = 5.67 \times 10^{15}$ eV^2: the 2.6% deviation is a numerical consistency check of two lattice normalizations (the $\chi_{\rm top}$ of [10] against the VMF anchor), and nothing more. The genuinely independent content of the target is the *value* of f_a itself, fixed by the neutrino sector.

2.5 Coupling to photons

The effective interaction is the standard axion-electrodynamics term [8]

$$\mathcal{L} \supset -\frac{g_{\theta\gamma\gamma}}{4}\,\theta\, F_{\mu\nu}\tilde{F}^{\mu\nu} = g_{\theta\gamma\gamma}\,\theta\,\mathbf{E}\cdot\mathbf{B}, \qquad g_{\theta\gamma\gamma} = \frac{\alpha}{2\pi}\,\frac{|E/N - 1.92|}{f_a}. \tag{9}$$

For the KSVZ benchmark [5] ($E/N = 0$) and the DFSZ benchmark [6] ($E/N = 8/3$):

$$g_{\theta\gamma\gamma}^{\rm KSVZ} = 2.08 \times 10^{-14}\ \text{GeV}^{-1}, \qquad g_{\theta\gamma\gamma}^{\rm DFSZ} = 8.10 \times 10^{-15}\ \text{GeV}^{-1}. \tag{10}$$

The model-dependent factor $|E/N - 1.92|$ is the only residual theory uncertainty on the coupling. We emphasize that $g_{\theta\gamma\gamma}$ is *not predicted* by NVG: the framework fixes f_a but not the UV anomaly ratio E/N, so KSVZ and DFSZ are benchmarks, not consequences of the model. A null result at KSVZ level therefore excludes only KSVZ-like realizations (Section 5.4). The KSVZ value is used as the discovery benchmark.

2.6 Relic abundance: the honest coherent fraction

For a post-inflationary scenario with the natural initial misalignment $\theta_i = \pi/\sqrt{3}$, the standard misalignment estimate [9] gives

$$\Omega_\theta h^2 \simeq 0.12\,\theta_i^2 \left(\frac{f_a}{10^{12}\ \text{GeV}}\right)^{7/6} = 0.0291, \tag{11}$$

i.e. **24% of the local dark-matter density** ($\Omega_{\rm DM}h^2 = 0.120$). The remaining density may reside in topological defects of the condensate sector; it does not contribute to the coherent haloscope signal. All signal powers below use the conservative coherent fraction $f_{\rm coh} = 0.24$; the $f_{\rm coh} = 1$ values are also quoted where relevant. A larger coherent fraction (e.g. from defect decay) would strengthen the signal proportionally.

2.7 Fifth force: the pseudoscalar null channel

The Compton range $\lambda_\theta = \hbar c/m_\theta = 3.72$ mm falls inside the sub-millimetre window of torsion-balance experiments [12]. Because θ is a *pseudo*scalar, the leading-order monopole–monopole Yukawa exchange between unmixed states vanishes, so unpolarized Eöt-Wash-class searches are expected to see nothing at this range. The residual channels — spin-dipole exchange with coupling m_N/f_a, loop-induced scalar admixtures, two-boson exchange, or condensate mixing — require a dedicated analysis and are not claimed here as quantitative predictions; we state only the robust null expectation for the unpolarized monopole channel.

3 Experimental Signature

The benchmark target is summarized in Table 1.

Table 1: Benchmark target of the NVG θ-DM sector, conditional on the θ-seesaw hypothesis with $c_\nu = 1$. No quantity in the right column is fitted to haloscope data.

Quantity	Predicted value	Origin
Decay constant f_a	1.07×10^{11} GeV	neutrino sector, Eq. (4)
Mass m_θ	53.2 μeV	Eq. (2), lattice χ_{top}
Frequency f_θ	12.86 GHz	$m_\theta c^2/h$
Virial linewidth Q_a	$\sim 10^6$	halo velocity dispersion
Line width Δf_a	12.8 kHz	f_θ/Q_a
$g_{\theta\gamma\gamma}$ (KSVZ)	2.08×10^{-14} GeV^{-1}	Eq. (9)
$g_{\theta\gamma\gamma}$ (DFSZ)	8.10×10^{-15} GeV^{-1}	Eq. (9)
Coherent DM fraction	24%	misalignment, Sec. 2.6
$m_\theta f_a$ vs QCD band	-2.6% deviation	consistency check

The signal is a nearly monochromatic microwave excess at 12.86 GHz with $Q_a \sim 10^6$ intrinsic quality factor, linearly polarized along the applied magnetic field, scaling as B_0^2, and tracking the sidereal modulation of the halo wind (a secondary discriminant).

4 Status of Published Searches

Table 2 compiles the published haloscope coverage around the predicted mass.

Table 2: Published haloscope mass windows around the NVG target. The benchmark point (53.2 μeV) lies in the unexplored gap between the two ORGAN windows, at the lower edge of the SMASH band (50–200 μeV) [11] that ORGAN declares as its roadmap target [16].

Experiment	Mass window (μeV)	Frequency (GHz)	Best limit on $g_{a\gamma\gamma}$ (GeV^{-1})
ADMX [14]	2.66–3.31	0.64–0.80	KSVZ reached
HAYSTAC II [15]	$\sim$17.3–19.5	4.18–4.71	$\sim 10^{-13}$
ORGAN [18]	25.45–26.27	6.15–6.35	2.8×10^{-13}
NVG target	**53.2**	**12.86**	$\mathbf{2.08 \times 10^{-14}}$
ORGAN [16]	63.2–67.1	15.3–16.2	3.0×10^{-12}
ORGAN 1b [17]	107.42–111.93	26.0–27.1	cogenesis ALP excluded
MADMAX [21, 24]	40–400 (design)	10–100	prototype phase

Two remarks. First, no published limit reaches $g = 2.1 \times 10^{-14}$ GeV^{-1} anywhere near 53 μeV: the closest result (ORGAN at 25.5–26.3 μeV) is a factor $\sim$13 weaker in coupling, at a different mass. Second, the ORGAN collaboration's stated goal is to scan the SMASH band 50–200 μeV under a constant magnetic field [16]; the NVG benchmark sits exactly at the lower edge of that band, making a targeted single-frequency run a natural addition to their program.

5 Test Protocol

5.1 Signal-power formula and its calibration

For a microwave cavity haloscope [8, 25],

$$P_{\text{sig}} = g_{\theta\gamma\gamma}^2 \frac{\rho_\theta}{m_\theta} B_0^2 \, V \, C \, \min(Q_L, Q_a), \tag{12}$$

where $\rho_\theta = f_{\rm coh}\,\rho_{\rm DM}$ with $\rho_{\rm DM} = 0.4$ GeV/cm^3, V is the cavity volume, C the mode form factor, and Q_L the loaded quality factor.

Calibration against published data. The ORGAN collaboration quotes, for its Phase 1a apparatus, $P_{\rm sig} \simeq 2.3 \times 10^{-25}$ W (KSVZ) and 3.1×10^{-26} W (DFSZ) for a 64 μeV axion [16]. Two checks:

1. *Apparatus-independent*: the coupling-coefficient ratio $(g_{\rm KSVZ}/g_{\rm DFSZ})^2 = (0.97/0.36)^2 = 7.26$ versus the published power ratio $2.3/0.31 = 7.42$ — agreement at the 2% level, confirming the normalization convention of Eq. (9).

2. *Absolute*: back-solving Eq. (12) from the published 2.3×10^{-25} W implies an apparatus product $VCQ \approx 0.44$ m^3 (for $B_0 = 7$ T), consistent with an ORGAN-scale cavity at 15.6 GHz with $Q \sim 10^4$–10^5. The unit chain of the present calculation is thus validated against a real haloscope at the factor-of-two level.

For the dielectric booster concept [19, 20, 21] the analogous expression is

$$P_{\rm sig} = g^2_{\theta\gamma\gamma}\,\rho_\theta\,B_0^2\,\frac{A\,\beta^2}{m_\theta^2}, \tag{13}$$

where A is the disk area and $\beta^2 \equiv P_{\rm sig}/P_0$ is the *power* boost relative to a bare magnetized mirror ($P_0 = g^2_{\theta\gamma\gamma}\rho_\theta B_0^2 A/m_\theta^2$), following the MADMAX convention [24] in which β itself denotes the amplitude boost; all boost figures quoted in this note are power boosts β^2. Both Eq. (12) and (13) have been re-derived within the Lorentz-reciprocity framework [25, 26]; an in-situ reciprocity measurement with a source antenna is the final normalization check of the stack before any dark-matter run.

5.2 Protocol A: resonant cavity

At $f_\theta = 12.86$ GHz the TM$_{010}$ mode requires a cylinder of radius $R = x_{01}c/(2\pi f_\theta) = 8.92$ mm. With length $L = 50$ mm, $V = 12.5$ cm^3, form factor $C = 0.69$, $B_0 = 10$ T and a superconducting (Nb) cavity with baseline $Q_L = 10^5$,

$$P_{\rm sig} = 6.3 \times 10^{-25}\ \text{W} \qquad (f_{\rm coh} = 0.24). \tag{14}$$

A single cavity reaches SNR $= 5$ only after $\sim$48 years: it is a pathfinder, not the discovery configuration. The viable cavity route is a phase-locked array (Table 3); $Q_L = 10^6$ at 13 GHz is flagged optimistic (Nb BCS losses scale as ω^2), so the $Q = 10^5$ rows are the engineering baseline.

Table 3: Integration time to SNR $= 5$ for cavity arrays ($B_0 = 10$ T, $T_{\rm sys} = 1$ K, $f_{\rm coh} = 0.24$, per-cavity signal from Eq. (12)).

	$N = 64$	$N = 256$
$Q_L = 10^5$ (baseline)	276 days	69 days
$Q_L = 10^6$ (optimistic)	0.3 days	0.1 days

5.3 Protocol B: fixed-position dielectric booster stack

The dielectric haloscope [19] replaces the resonant cavity by $\sim$80 dielectric disks (LaAlO$_3$, $\varepsilon \approx 24$) in a strong magnetic field; constructive interference of the emitted photons provides the power boost β^2. In a scanning experiment the disk positions must be moved mechanically — the technically hardest subsystem, now being commissioned [23]. **The point-frequency target**

eliminates this subsystem entirely: at a known frequency the inter-disk spacing is fixed, and the stack becomes a static optical element.

With $D = 1.2$ m disks ($A = 1.13$ m^2, the low-mass-end MADMAX aperture), $B_0 = 10$ T and $f_{\rm coh} = 0.24$, Eq. (13) gives (Table 4). The power-boost targets $\beta^2 = 10^4$–10^5 are the design goals of the MADMAX program [21]; the three-disk prototype has so far demonstrated β^2 up to ∼2000 with 13–17% uncertainty [24], which we state as the principal technical risk of this option.

Table 4: Fixed dielectric stack, $B_0 = 10$ T, $T_{\rm sys} = 1$ K, readout bandwidth equal to the axion linewidth $\Delta f_a = 12.8$ kHz (the stack boost band $\gg \Delta f_a$).

Power boost β^2	$P_{\rm sig}$ ($f_{\rm coh} = 0.24$)	$P_{\rm sig}$ ($f_{\rm coh} = 1$)	t(SNR = 5)
10^4	3.1×10^{-23} W	1.3×10^{-22} W	18 h
10^5	3.1×10^{-22} W	1.3×10^{-21} W	11 min

Protocol B is strictly simpler than the cavity array: no cryogenic tuning mechanism, no phase-locked multi-cavity readout, a single wide-bore magnet, horn antenna and a near-quantum-limited amplifier — while the required power boost β^2 is already the stated MADMAX design target in a frequency band that includes 12.86 GHz.

5.4 Readout, statistics, and preregistered criteria

Noise. $T_{\rm sys} \approx 1$ K is adopted: millikelvin physical temperature plus a flux-driven Josephson parametric amplifier. This is not an assumption but demonstrated technology — ORGAN's most recent phase operates exactly this way [18].

Statistical target definition. With the Dicke radiometer relation

$$\mathrm{SNR} = \frac{P_{\rm sig}}{k_B T_{\rm sys}} \sqrt{\frac{t}{\Delta f}}, \tag{15}$$

the narrow search window below is conditional on the benchmark coefficient $c_\nu = 1$. Statistical uncertainties in m_3 and $\chi_{\rm top}$ are propagated by Monte Carlo (Appendix 7: $m_3 \sim \mathcal{N}(50.3, 0.25)$ meV, $\chi_{\rm top}^{1/4} \sim \mathcal{N}(75.5, 0.7\%)$ MeV), while c_ν is *not* assigned a statistical prior: it is an explicit theory parameter. Broader intervals obtained by varying c_ν below are sensitivity scenarios, not confidence intervals.

- **Conditional benchmark window ($c_\nu = 1$ fixed).** The target-frequency distribution has median 12.86 GHz and a 68% interval [12.67, 13.05] GHz (relative width 1.5%, dominated by $\chi_{\rm top}$). The corresponding pre-specified 3σ analysis window is [12.29, 13.43] GHz, resolved in bins of width Δf_a.

- **Illustrative coefficient scenario (c_ν varied over [1/2, 2]).** Via Eq. (6) this non-statistical sensitivity scenario broadens the target range to 6.4–25.7 GHz. It is a scan, not a point-target test, and must carry its full trials factor and sensitivity variation across the band.

- **Extreme envelope ($c_\nu \in [1/3, 3]$).** $f_\theta \in [4.3, 38.6]$ GHz (∼9 times the benchmark band): the full scan cost if c_ν is never fixed by theory.

Signal power and sensitivity across the band. Along the curve (6), *at fixed coherent fraction*, the signal powers scale simply: $P_{\rm sig}^{\rm stack} \propto g^2/m_\theta^2 \propto c_\nu^0$ for the dielectric stack (exactly invariant under c_ν rescaling) and $P_{\rm sig}^{\rm cavity} \propto g^2/m_\theta \propto c_\nu^{-1}$ for the cavity, so the integration times of

Tables 3–4 apply unchanged to the stack and rescale as c_ν^2 for the cavity at any other benchmark value of c_ν. If instead the misalignment estimate of Sec. 2.6 is imposed, the coherent fraction itself scales as $f_{\rm coh} \propto f_a^{7/6} \propto c_\nu^{7/6}$, which multiplies both powers accordingly. Across coefficient scenarios the required sensitivity is therefore not constant: at fixed E/N, $g_{\theta\gamma\gamma} \propto c_\nu^{-1}$, while the assumed misalignment abundance also changes with $f_a \propto c_\nu$; consequently, a broad-c_ν program requires a frequency-dependent sensitivity forecast rather than a uniform scan-time rescaling. The $\pm 0.5\%$ window of the v1 draft covered only m_3 and $\chi_{\rm top}$ and is superseded.

Trials factor and global significance. The benchmark prediction is a *conditional narrow target* in model-parameter space, but the experimentally searched frequency interval still contains many statistically independent axion-line templates: the 1.14 GHz benchmark window holds $N_{\rm eff} \simeq 8.9 \times 10^4$ independent bins of width $\Delta f_a = 12.8$ kHz. A local 5σ excess scanned over the whole window corresponds to only $\sim 2\sigma$ globally, so the discovery statistic must report both a local significance and a global significance corrected for the pre-specified $N_{\rm eff}$. Pre-registration fixes the analysis window and procedure; it does not remove the trials correction.

Discovery criterion. A candidate must exceed a pre-specified local threshold corresponding to a *global* 5σ false-alarm probability after correction for $N_{\rm eff}$ templates in the benchmark window. For the conservative independent-bin estimate above, $N_{\rm eff} \simeq 8.9 \times 10^4$ implies a local threshold of approximately 6.9σ. The final $N_{\rm eff}$ shall be obtained from the matched-filter bank and verified with noise-only Monte Carlo. The candidate must also persist under independent retaking, scale as B_0^2 when the magnet current is varied, match the expected Maxwell–Boltzmann halo lineshape, and be reproduced at a second apparatus with a different magnet before any claim.

Exclusion criterion. Exclusion is reported at two levels. (1) A null result at KSVZ-level sensitivity, $g_{\theta\gamma\gamma} \leq 2.1 \times 10^{-14}$ GeV^{-1} across the preregistered window, excludes the *KSVZ-benchmark realization* of the θ-seesaw target at $f_{\rm coh} = 0.24$ — and only that realization, since E/N is UV-dependent. (2) A null at DFSZ level (8.1×10^{-15} GeV^{-1}) additionally covers the DFSZ benchmark; excluding the neutrino–axion linkage itself would further require a fixed E/N assignment and a cosmological determination of $f_{\rm coh}$. Because exclusion curves assume the full local density, the effective bound on g for a coherent fraction $f_{\rm coh}$ is weaker by $1/\sqrt{f_{\rm coh}} \approx 2.0$; this rescaling must be applied when interpreting any null.

Side-channel cross-checks. The same f_a predicts $\sum m_\nu = 59$ meV (CMB+BAO, ongoing) and a $0\nu\beta\beta$ window $m_{\beta\beta} \in [0, 6.4]$ meV (LEGEND-1000, nEXO): these provide framework-level corroboration independent of haloscopes.

6 Honest Caveats

1. **Framework hypothesis and its coefficient.** The θ-seesaw, Eq. (3), is a postulate of the NVG construction, and its UV origin is not derived. The coefficient c_ν — encoding the two-loop anomaly diagram, possible flavor factors and RG running — is unknown and is propagated as a parameter throughout; $c_\nu = 1$ is a benchmark, not a derived value. If Eq. (4) fails — e.g. m_3 from a future $0\nu\beta\beta$/oscillation combination disagrees with 50.3 meV — the entire frequency target moves. The haloscope test is precisely the discriminator.

2. **Spectral premise.** The input $m_3 = 50.3$ meV assumes normal ordering and a near-minimal lightest neutrino mass. Oscillation experiments measure only mass-squared differences, and cosmology bounds the sum of masses without fixing m_3 at this precision. The target is therefore conditional on this premise; the Monte Carlo window of Sec. 5.4

covers the stated uncertainty, and a shift to inverted ordering or a heavier lightest state would move the target outside the benchmark window.

3. **Model dependence of the coupling.** The factor $|E/N - 1.92|$ in Eq. (9) is benchmarked to KSVZ/DFSZ; a different UV completion rescales $g_{\theta\gamma\gamma}$ without moving the frequency. The mass target is unaffected; the coupling is not predicted.

4. **Coherent fraction.** All powers use $f_{\rm coh} = 0.24$ (misalignment only). If most of the sector's density sits in incoherent defects, haloscope signals weaken accordingly; if defects decay into coherent modes, they strengthen. This is the dominant astrophysical uncertainty and is why the exclusion criterion carries the $1/\sqrt{f_{\rm coh}}$ rescaling.

5. **Technology risk.** $Q_L = 10^6$ at 13 GHz and $\beta^2 = 10^5$ are program goals, not demonstrated hardware (the MADMAX prototype reached $\beta^2 \sim 2000$). The baseline rows ($Q_L = 10^5$, $\beta^2 = 10^4$) reflect the validated state of the art.

7 Conclusion

This note is a phenomenological target paper. Under the stated θ-seesaw hypothesis with benchmark coefficient $c_\nu = 1$, a single neutrino-sector input of the NVG/VMF framework fixes $f_a = 1.07 \times 10^{11}$ GeV, which through lattice-QCD physics yields a conditional point target: $m_\theta = 53.2\ \mu$eV, a microwave line at 12.86 GHz with KSVZ-level photon coupling. The point lands on the QCD axion band within 2.6% (a normalization cross-check, not an independent test), sits in the currently unexplored gap between published ORGAN windows, and is directly on that collaboration's declared SMASH roadmap. Because the target is a point, the search needs no scanning: a fixed dielectric booster stack reaches discovery-level significance in under a day of integration at design power boost $\beta^2 = 10^4$. The protocol above is preregistered, the frequency window is derived by Monte Carlo from all stated uncertainties, and the exclusion statements are scoped precisely: a KSVZ-level null closes the KSVZ-benchmark realization of the target, not every conceivable θ-model. Either outcome is informative — which is the property a target must have to deserve an experiment.

Appendix A: Reproducibility

All numbers in this note are produced by scripts in the accompanying repository (`https://github.com/infosave2007/vmf`, Zenodo-archived releases):

- `generator_project/calc_theta_haloscope_design.py` (v4) — geometry, signal powers (Eqs. (12)–(13), MADMAX β^2 convention), ORGAN calibration, integration times, published-search status, the c_ν-parametrized θ-seesaw chain, and the Monte Carlo block that produces the preregistered window of Sec. 5.4;

- `generator_project/calc_theta_haloscope_v2.py` — frequency derivation, axion-band consistency check (2.6%);

- `verification/nvg_neutrino_seesaw.py` — Eq. (4), fixing f_a; `verification/nvg_theta_fifth_force.py` — pseudoscalar null channel.

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